PROJECT REPORT OF CO-OP Project at Industry (Module-2)

ON

<u>AI-Driven Demand Forecasting Model for Micro-Retailers Using Machine Learning</u>

submitted in partial fulfilment of the requirements for the award of degree of

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DECLARATION

We hereby certify that the work which is being presented in the project report entitled **“AI-Driven Demand Forecasting Model for Micro-Retailers Using Machine Learning”** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the Department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of our own work carried out under the supervision of **Ms. Simarjit Kaur**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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ABSTRACT

Micro-retailers, popularly known as kirana stores, play an important role in serving India's vast retail ecosystem. Although these micro-retailers contribute significantly to the economy, most of them operate without formal records and rely on intuition-based purchasing decisions. With the widespread adoption of affordable digital billing and UPI-based transactions, transactional data is increasingly available, creating a viable opportunity to employ Artificial Intelligence (AI) in demand forecasting for micro-retailers.

This project develops an AI-driven framework for short-term demand prediction in micro-retail outlets aimed at enhancing inventory planning accuracy, reducing stockouts, and minimizing misutilization of working capital. Machine learning algorithms — including Moving Average, ARIMA, Random Forest Regressor, and Long Short-Term Memory (LSTM) networks — were developed and compared. Data processing incorporated the effects of seasonality, festivals, and weekly demand trends. Performance was evaluated using Mean Absolute Error (MAE) and Root Mean Square Error (RMSE).

Experimental results across five kirana stores over an 18-month observation window demonstrate that the LSTM-based model achieves the best prediction accuracy, reducing forecasting errors by approximately 18 to 25 percent compared with classical baselines, particularly excelling at capturing seasonal spikes and festival-driven demand variations. This study establishes the viability of AI-based forecasting to maximize inventory efficiency for micro-retailers, reduce operational losses, and advance the digital agenda in the unorganized retail sector.

Keywords: demand forecasting, machine learning, micro-retail, inventory optimization, LSTM, ARIMA, Random Forest, kirana stores.

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1. Introduction

1.1 Background

Demand forecasting is the process of estimating future customer demand for products by analysing historical sales data, trends, and external influencing factors. It plays a crucial role in inventory management, supply chain planning, and pricing. Large retail chains leverage advanced analytics and machine learning techniques that enable exponential business growth; however, micro-retailers often lack access to such capabilities.

In the retail ecosystem, micro-retailers — locally known as kirana stores in India — form a significant segment, particularly in developing economies. These stores operate with limited resources, minimal technology, and low technical expertise. They rarely perform complex analytics on historical sales data and instead rely on manual or experience-based decision-making for inventory management. As a result, overstocking, understocking, and demand fluctuations are common, directly impacting profitability and customer satisfaction.

1.2 Problem Statement

Traditional demand forecasting methods, including rule-based approaches and simple statistical models, are often insufficient in micro-retail environments due to the complexity and variability of demand patterns. Factors such as irregular purchasing behaviour, seasonal variations, local events, and external influences like weather contribute to highly dynamic demand, making accurate predictions a challenging task.

Compounding this difficulty is the data sparsity inherent to small-format stores. Historical records are often incomplete, inconsistently captured, and limited to a small number of SKUs with reliable sales history. Existing forecasting solutions, designed for organised retail with abundant clean data, do not transfer well to this setting. There is a clear need for a forecasting framework specifically engineered for the data, infrastructure, and operational realities of micro-retail.

1.3 Objectives

The principal objectives of this project are summarised below.

- To design an end-to-end demand forecasting framework tailored to the operational realities of Indian micro-retail (kirana) stores.
- To compare a representative statistical baseline (ARIMA), a classical machine-learning model (Random Forest Regressor), and a deep-learning model (LSTM) on real kirana sales data.

- To engineer features that capture lag dependencies, weekly cycles, festival effects, and broader seasonality from raw transactional data.
- To address practical issues of sparse demand and the cold-start problem for newly introduced SKUs.
- To quantify forecasting improvements using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE).
- To demonstrate that data-driven inventory intelligence is feasible for small-format retailers, not just large organised retail chains.

1.4 Scope of the Project

The scope of this project is restricted to short-term (one-week-ahead) demand forecasting for individual SKUs sold through digitally enabled kirana stores. The empirical study uses 18 months of UPI and digital-billing transaction data drawn from five stores in urban and peri-urban Punjab. External signals such as weather and social-media trends, while acknowledged as potentially useful, are out of scope and earmarked for future work. The framework is designed to be deployable on commodity hardware so that forecasts can ultimately be served from a lightweight mobile or desktop interface accessible to a non-technical store owner.

1.5 Organization of the Report

The remainder of this report is organised as follows. Chapter 2 describes the proposed methodology, including the four-stage pipeline of data collection, feature engineering, model training, and inference. Chapter 3 details the tools, libraries, and hardware used in the implementation. Chapter 4 covers the experimental setup and the implementation specifics of each forecasting model. Chapter 5 presents and discusses the empirical results across overall, festival-period, and cold-start scenarios. Chapter 6 concludes with a summary of findings, practical implications, limitations, and directions for future work, followed by References and Appendices.

2. Methodology

2.1 Overview of the Proposed Approach

The proposed framework follows a four-stage pipeline tailored for the micro-retail environment: (i) data collection and preprocessing, (ii) feature engineering, (iii) model training and selection, and (iv) inference with business output. Figure 2.1 illustrates the overall flow.

Pipeline: Raw Sales Data → Preprocessing → Feature Engineering → Model Training → Demand Forecast

Fig. 2.1 Proposed AI-driven demand forecasting pipeline for micro-retailers.

2.2 Data Collection and Preprocessing

Raw sales data sourced from digital billing and UPI transaction logs are inherently noisy, incomplete, and inconsistently recorded in micro-retail settings. The preprocessing stage first handles missing values through zero-filling for confirmed no-sale days and mean substitution for gaps caused by system outages. Duplicate and irrelevant entries are removed to maintain data reliability.

Sales records are subsequently aggregated into a time-series format at daily and weekly granularity. Daily aggregation preserves intra-week purchasing patterns, while weekly aggregation reduces noise and improves pattern visibility for slower-moving SKUs. Finally, min-max normalisation is applied to scale all features to the [0, 1] interval, preventing large-valued variables from disproportionately influencing the model during training.

2.3 Feature Engineering

To capture the factors driving product demand, a rich feature set is constructed from both temporal and contextual signals.

- Lag features: historical sales values at lag intervals of 1, 7, and 14 days provide autoregressive signals.
- Temporal features: day-of-week (encoded as cyclic sine–cosine pairs), week-of-month, and month-of-year capture periodic purchasing patterns.
- Festival indicators: binary variables flag major festival periods (Diwali, Holi, Eid, Christmas), public holidays, and notable local events.
- Seasonality index: derived using Fourier decomposition of the training set to capture annual demand cycles.

- Cold-start proxies: for newly added SKUs, category-level sales aggregates serve as features until sufficient product-level history is accumulated.

2.4 Model Selection and Architecture

Three forecasting models are developed and compared within the proposed framework:

- ARIMA — a statistical baseline configured with automatically selected (p, d, q) parameters using the Akaike Information Criterion (AIC).
- Random Forest Regressor — a classical machine-learning baseline employing 300 estimators with a maximum depth of 15, trained on the full engineered feature set.
- LSTM — the primary deep-learning model, consisting of two stacked LSTM layers of 64 and 32 units respectively, followed by a fully connected Dense layer with linear activation. Training uses the Adam optimiser with a learning rate of 0.001, a batch size of 32, and up to 100 epochs with early stopping based on validation loss.

Model selection follows a structured evaluation protocol where all models are trained on 70% of the data, validated on 15%, and tested on the held-out 15%. Hyperparameters are tuned per model: randomised search for Random Forest, AIC-based grid search for ARIMA, and Bayesian optimisation for LSTM.

2.5 Handling Data Scarcity and Cold-Start

For products with fewer than 30 days of sales history, category-level aggregated forecasts are used as the demand estimate until the product accumulates sufficient history. Transfer learning is applied for mid-frequency SKUs by initialising LSTM weights from a pre-trained model on high-frequency product data, and then fine-tuning on the target SKU. This strategy significantly reduces cold-start forecasting error compared with training from scratch and allows the framework to serve newly introduced products from day one.

3. Tools and Technologies

3.1 Programming Language and Core Libraries

The entire pipeline is implemented in Python 3.10, chosen for its mature ecosystem in machine learning and time-series analysis. The following core libraries are used throughout the project.

- NumPy and pandas for numerical computation and tabular data manipulation.
- scikit-learn 1.3 for the Random Forest Regressor, train/validation/test splits, randomised hyperparameter search, and standard evaluation metrics.
- statsmodels 0.14 for the ARIMA implementation and AIC-based order selection.
- TensorFlow 2.12 with the Keras high-level API for building, training, and evaluating the LSTM network.
- Matplotlib and Seaborn for exploratory data analysis and the figures presented in Chapter 5.
- Optuna for Bayesian hyperparameter optimisation of the LSTM model.

3.2 Hardware Setup

All experiments are conducted on a workstation equipped with an Intel Core i7-12700 CPU and 32 GB RAM. The LSTM model is additionally accelerated using an NVIDIA RTX 3060 GPU with 12 GB of VRAM, which materially reduces the time per training epoch and makes Bayesian hyperparameter search feasible within the project timeline. ARIMA and Random Forest are trained on CPU.

3.3 Development Environment

Development is performed inside Jupyter Notebook for exploratory work and Visual Studio Code for the production-grade pipeline. Version control is handled through Git with the project repository hosted privately. Virtual environments are managed using venv to ensure reproducibility, and a requirements.txt file pins the exact versions of all dependencies.

3.4 Data Sources

Primary data are drawn from the digital billing software and UPI payment logs of five participating kirana stores in Punjab, exported as CSV files at the end of each weekly observation cycle. Secondary contextual data include the official list of Indian public holidays and the dates of major regional festivals during the observation window, sourced from publicly available calendars.

4. Implementation

4.1 Dataset Description

The experimental dataset comprises 18 months of daily sales records collected from five micro-retail kirana stores located in urban and peri-urban areas of Punjab, India. The dataset covers 312 unique Stock Keeping Units (SKUs) across seven product categories: groceries, dairy, beverages, personal care, household goods, snacks, and staples. In total, the dataset contains approximately 1.2 million transaction records after deduplication and cleaning.

The collection period spans two major festival seasons — Diwali 2022 and Holi 2023 — enabling evaluation of model performance under significant demand-spike conditions. Stores adopted UPI-based digital billing systems six to twelve months prior to the start of the data collection window, ensuring that sufficient digital records were available for model training.

4.2 Data Preprocessing Pipeline

Each store's raw export is loaded as a separate pandas DataFrame, parsed for date and time, and joined against a master SKU table to ensure consistent product identifiers across stores. Missing values are imputed according to the rules described in Chapter 2. Outlier transactions, defined as units sold per day exceeding the 99.5th percentile per SKU, are winsorised to limit their leverage during model fitting. Finally, all features are scaled with min-max normalisation, and the cleaned data are stored as Parquet files for efficient downstream loading.

4.3 Model Training

Each forecasting model is trained per SKU on the chronologically ordered training window. For ARIMA, the (p, d, q) tuple is selected automatically using AIC minimisation over a bounded grid. The Random Forest Regressor is fit to the engineered feature matrix, with hyperparameters (number of trees, maximum depth, minimum samples per leaf) tuned through 50 iterations of randomised search. The LSTM is trained for up to 100 epochs with early stopping (patience of 10 epochs) on the validation MAE; layer sizes, dropout rate, and learning rate are tuned through 30 trials of Bayesian optimisation.

4.4 Cross-Validation Strategy

To produce unbiased performance estimates that respect the temporal structure of the data, a rolling-window cross-validation scheme is employed. The window size is 90 days, the prediction horizon is 7 days (one-week-ahead forecast), and the window is advanced by 7 days at each fold.

This procedure is repeated across the full 18-month observation window for every SKU, and the per-fold errors are averaged to produce the metrics reported in Chapter 5.

4.5 Evaluation Metrics

Model performance is quantified using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE). MAE measures the average magnitude of forecast errors. RMSE penalises large errors more heavily due to the squared term, making it sensitive to outlier demand events. MAPE expresses error as a percentage of actual demand, providing a scale-independent measure that is useful for comparing across SKUs with very different sales volumes.

$$MAE = (1 / n) \times \Sigma \mid y_i - \hat{y}_i \mid$$

$$RMSE = \sqrt{[(1 / n) \times \Sigma (y_i - \hat{y}_i)^2]}$$

$$MAPE = (100 / n) \times \Sigma \mid (y_i - \hat{y}_i) / y_i \mid$$

5. Major Findings, Outcomes and Results

5.1 Overall Forecasting Performance

Table 5.1 summarises the mean performance of the three models across all 312 SKUs and all five stores over the 18-month evaluation period. The LSTM-based model achieves the lowest error across all three metrics, confirming its superior ability to capture complex temporal patterns in micro-retail demand data.

Table 5.1 Overall model performance across all 312 SKUs (18-month average).

Model	MAE	RMSE	MAPE (%)
ARIMA	5.84	8.43	19.7
Random Forest	4.01	5.94	14.2
LSTM (Proposed)	3.21	4.87	11.4

The LSTM model attains a MAE of 3.21 units, an RMSE of 4.87 units, and a MAPE of 11.4%, representing error reductions of approximately 18 to 25 percent relative to the ARIMA baseline and 10 to 14 percent relative to Random Forest. ARIMA, while straightforward to implement, struggles with the non-linear and irregular demand patterns characteristic of micro-retail, particularly during festival periods. Random Forest performs considerably better than ARIMA owing to the richness of the engineered feature set, although it does not match LSTM accuracy on high-spike events.

5.2 Performance During Festival Periods

During the Diwali 2022 window, a five-day period of substantially elevated demand, the LSTM model reduced RMSE by 31 percent relative to ARIMA and 17 percent relative to Random Forest. Demand spikes of 2.5x to 4.2x baseline daily sales were observed across snack, beverage, and gift-item categories, and LSTM forecasts tracked these spikes with notably lower lag than the comparison models. This improvement is attributed to the LSTM's ability to combine multi-step temporal context with the binary festival indicator features introduced during feature engineering.

Table 5.2 Model performance during Diwali 2022 festival window (5-day period).

Model	MAE	RMSE	MAPE (%)
ARIMA	9.12	13.47	28.3
Random Forest	6.34	9.11	18.6
LSTM (Proposed)	4.38	6.42	13.1

During non-festival weeks, the performance gap between LSTM and Random Forest narrows, with both models achieving comparable MAPE values in the 9 to 13 percent range. This suggests that for routine inventory planning, a lighter-weight Random Forest model may be a practical alternative when computational infrastructure is limited, as it offers reasonable accuracy with significantly lower training and inference time.

5.3 Cold-Start and Data Scarcity Results

For newly added SKUs with fewer than 30 days of history, the category-level proxy forecasting strategy reduced the initial MAPE from an unconstrained model estimate of approximately 41 percent to 22 percent — a substantial improvement. Once sufficient product-level history accumulated, beyond 45 days, fine-tuned LSTM models further reduced MAPE to 11.8 percent, approaching the performance achieved for established high-frequency SKUs.

Table 5.3 Cold-start MAPE reduction by forecasting strategy.

Strategy	MAPE (<30 days)	MAPE (>45 days)
No strategy (baseline)	41%	20%
Category-level proxy	22%	15%
Fine-tuned LSTM	—	11.8%

5.4 Discussion

The empirical findings collectively support the viability of AI-driven demand forecasting in the micro-retail context. Three observations stand out. First, sequence-aware deep learning materially improves accuracy over both statistical and tabular machine-learning baselines, particularly when demand is non-stationary or driven by exogenous events such as festivals. Second, careful feature engineering — especially the inclusion of festival indicators and lag features — remains essential even when using deep learning, and contributes a non-trivial share of the overall accuracy gain. Third, practical interventions for the cold-start problem are not optional but central; without them, model performance for newly introduced SKUs would be too poor to support actionable inventory decisions.

Compared with the retailer's intuition-based ordering baseline, which exhibits an estimated MAPE of approximately 32 percent based on retailer-reported stockout and overstock incidents, even the ARIMA model substantially improves inventory accuracy, and the LSTM model nearly triples it. Translated into operational terms, this level of accuracy improvement is sufficient to meaningfully reduce both stockout-driven lost sales and overstock-driven working-capital lock-in for participating stores.

6. Conclusion and Future Scope

6.1 Summary of Findings

This project has presented an AI-driven demand forecasting framework specifically designed for the micro-retail environment, addressing the unique challenges of data scarcity, sparse and irregular demand patterns, infrastructure constraints, and heterogeneity across stores and products. The proposed framework integrates robust preprocessing, context-aware feature engineering, and a comparative evaluation of three forecasting models: ARIMA, Random Forest, and LSTM.

Experimental results across five kirana stores over 18 months demonstrate that the LSTM-based model consistently outperforms both statistical and classical machine-learning baselines, achieving forecasting error reductions of 18 to 25 percent over ARIMA and yielding the strongest performance during festival-driven demand spikes. The Random Forest model provides a practical alternative for resource-constrained deployments where deep-learning infrastructure is unavailable.

6.2 Practical Implications

The study establishes that AI-based demand forecasting is both feasible and beneficial for micro-retailers in India, with the potential to significantly reduce stockouts, lower excess inventory costs, and improve overall operational efficiency. By leveraging digital transaction data increasingly available through UPI and digital billing adoption, even small-scale retailers can access data-driven inventory intelligence previously reserved for large organised retail chains. For an individual kirana owner, this translates into more accurate weekly purchase orders, fewer instances of fast-moving items being out of stock around festivals, and lower working capital tied up in slow-moving inventory.

6.3 Limitations

The study has several limitations that should be acknowledged. First, the empirical evaluation is restricted to five stores in a single geographic region, and the generalisability of the results to other regions, store formats, or product mixes remains to be validated. Second, the dataset, while substantial, spans only 18 months and may not fully capture multi-year cyclical effects. Third, exogenous factors such as weather, local economic activity, and social-media-driven consumption trends are not yet incorporated, and may further improve accuracy if integrated. Finally, the LSTM model requires GPU support for efficient training, which limits on-premise deployment in stores with very basic hardware.

6.4 Future Work

Several promising extensions of this work are planned.

- Integrate external contextual data sources such as weather forecasts and social-media trend signals to capture demand drivers beyond historical sales.
- Develop a lightweight, mobile-friendly inference interface so that kirana store owners can directly consume forecasts and recommended order quantities.
- Extend the framework to multi-store collaborative forecasting, in which knowledge gained from one store accelerates learning for similar stores in the same cluster.
- Investigate explainability techniques (such as SHAP or attention-based visualisations) so that store owners can understand and trust the system's recommendations.
- Extend the evaluation to additional regions and store formats to test generalisability beyond the current five-store sample.

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APPENDICES

Appendix A — Hyperparameter Configurations

ARIMA: order (p, d, q) selected per SKU using AIC minimisation over $p \in \{0 \ldots 5\}$, $d \in \{0, 1, 2\}$, $q \in \{0 \ldots 5\}$. Random Forest: 300 estimators, maximum depth 15, minimum samples per leaf 2, minimum samples per split 5, max_features = sqrt. LSTM: two stacked LSTM layers (64 and 32 units), dropout 0.2 between layers, Adam optimiser (learning rate 0.001), batch size 32, up to 100 epochs with early stopping (patience 10) on validation MAE.

Appendix B — Feature List (Engineered)

Lag features: sales_lag_1, sales_lag_7, sales_lag_14. Temporal features: dow_sin, dow_cos, week_of_month, month, is_month_start, is_month_end. Festival features: is_diwali, is_holi, is_eid, is_christmas, is_public_holiday, is_local_event. Seasonality features: fourier_1_sin, fourier_1_cos, fourier_2_sin, fourier_2_cos. Cold-start features: category_avg_sales_7d, category_avg_sales_30d.

Appendix C — Dataset Summary by Store

Store 1 (Rajpura, urban): 78 SKUs, ~245k transactions. Store 2 (Patiala, urban): 84 SKUs, ~260k transactions. Store 3 (Mohali, peri-urban): 64 SKUs, ~210k transactions. Store 4 (Banur, peri-urban): 51 SKUs, ~180k transactions. Store 5 (Kharar, peri-urban): 62 SKUs, ~205k transactions. Total: 312 unique SKUs (after deduplication across stores), approximately 1.2 million cleaned transaction records.